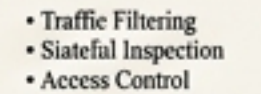
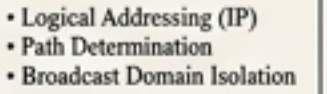
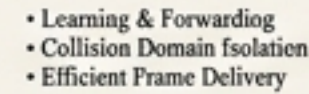
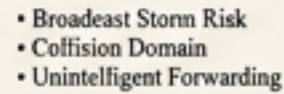
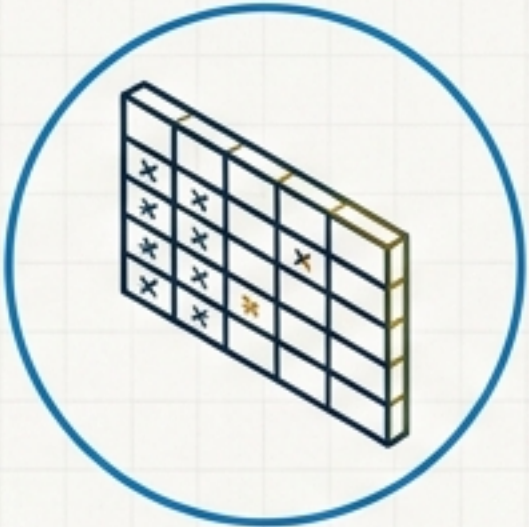
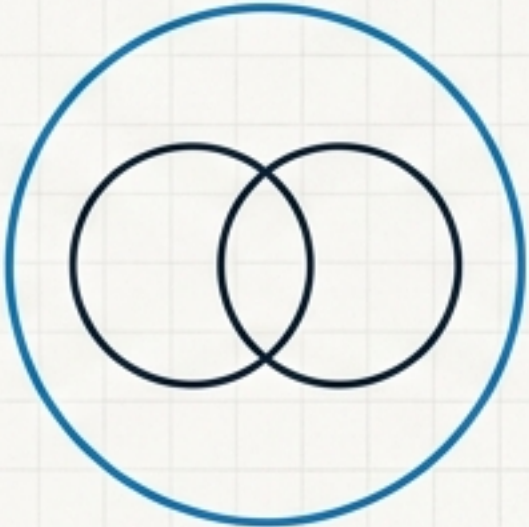


Decoding the logical behaviors of Hubs, Switches, Routers, and Firewalls.



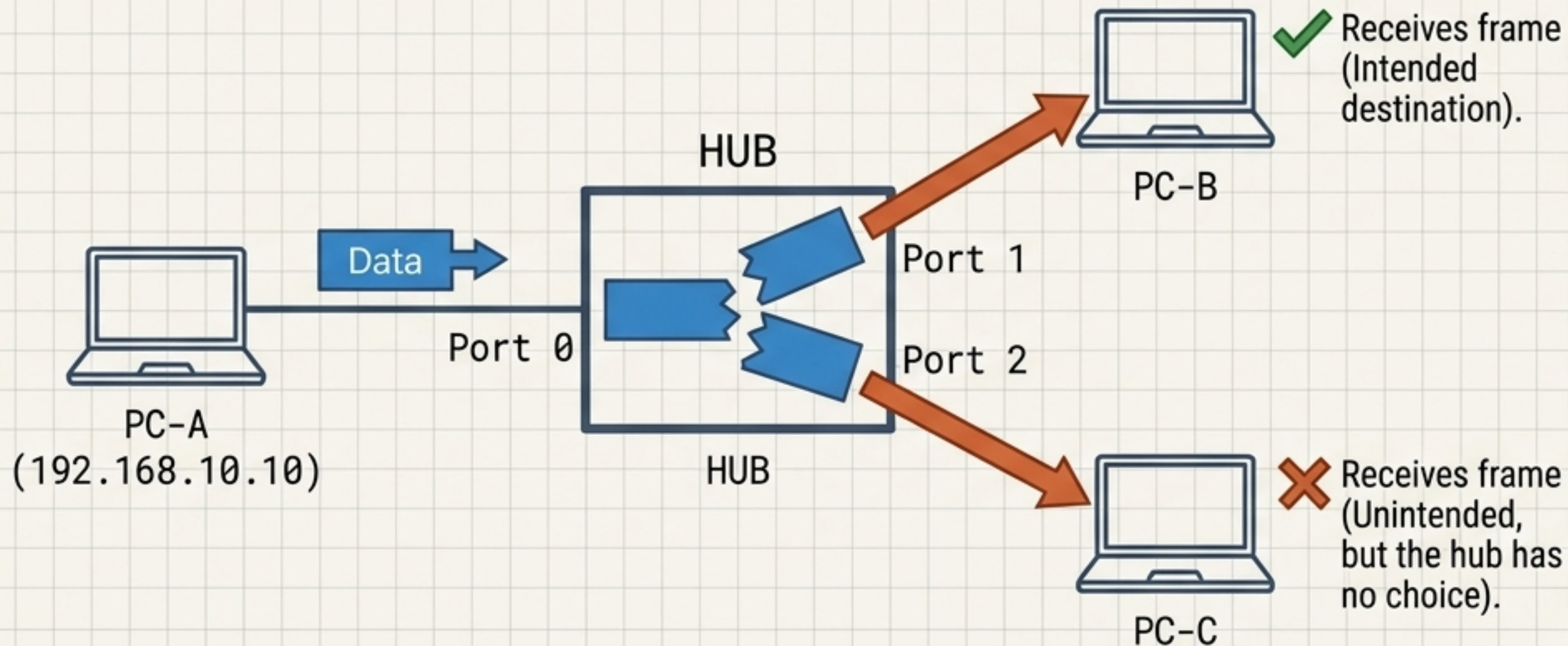
THE UNIVERSAL DEVICE FRAMEWORK

To understand any network device, do not memorize its name.
Instead, decode its behavior by answering five foundational questions.

				
1. Operating Layer What OSI layer does it primarily operate at? ▪ Layer 1 ▪ Layer 2 ▪ Layer 3 ▪ Multiple	2. Decision Info What information does it read to make decisions? ▪ Physical signal ▪ MAC address ▪ IP address ▪ Policy rules	3. Memory Table What table does it maintain in memory? ▪ None ▪ MAC / CAM Table ▪ Routing Table ▪ ACL / Rule Set	4. Domain Behavior How does it handle collision and broadcast domains? ▪ Shared domains ▪ Separated domains	5. Forwarding Rule What is its core forwarding logic? ▪ Repeat blindly ▪ Match MAC to port ▪ Route via IP map ▪ Allow / Deny

Layer 1: The Hub (Physical Repeater)

The Floodlight



DIAGNOSTICS

Layer: 1

Decision Info:
Physical electrical signal
(No frame awareness)

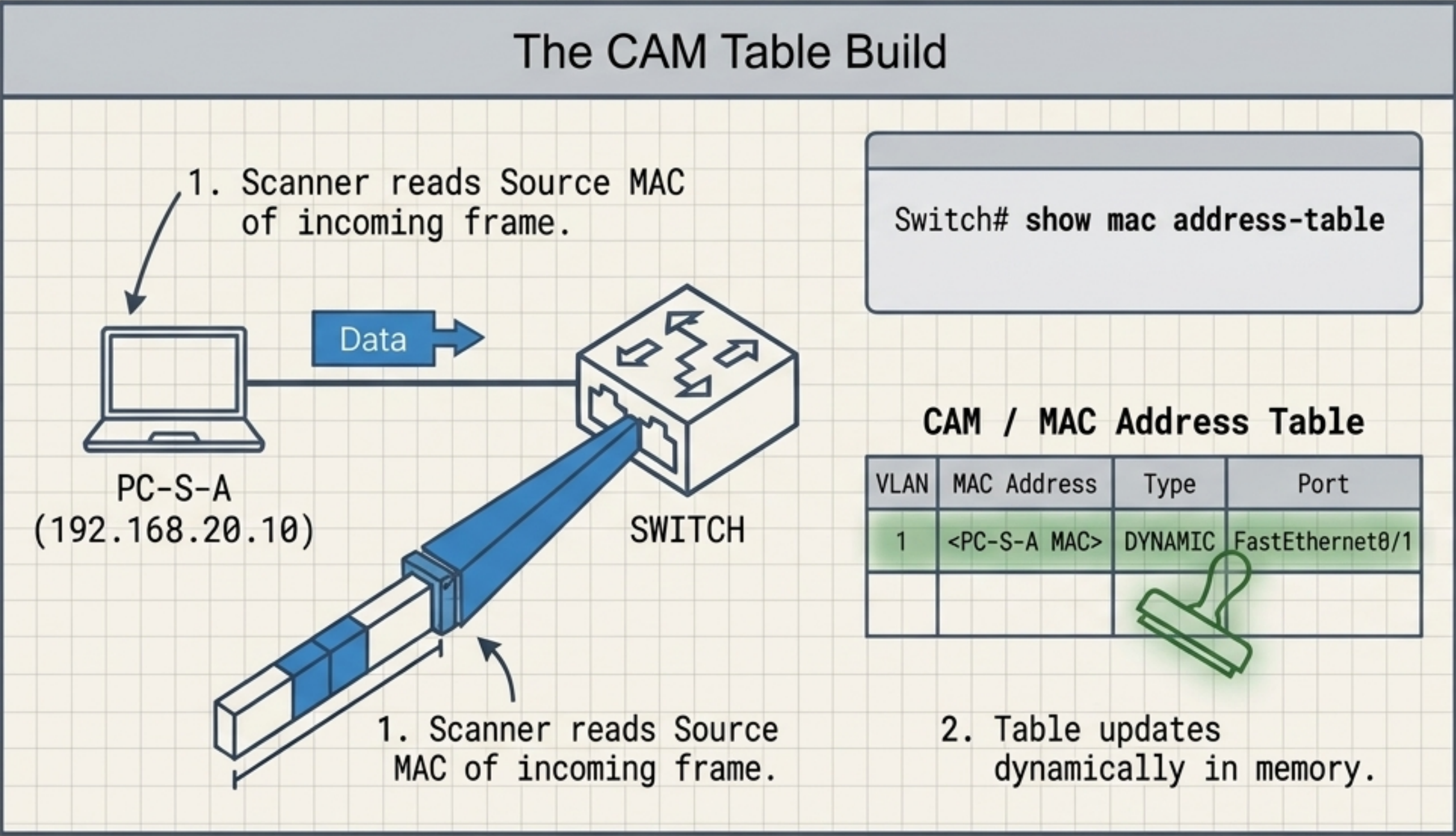
Table Maintained:
None

Domains:
1 shared collision domain

Forwarding Rule:
Repeat blindly to all ports

Layer 2: The Switch (MAC Learning)

A switch is not a smarter hub. It possesses memory. It learns network geography automatically by reading the source MAC address of incoming frames.



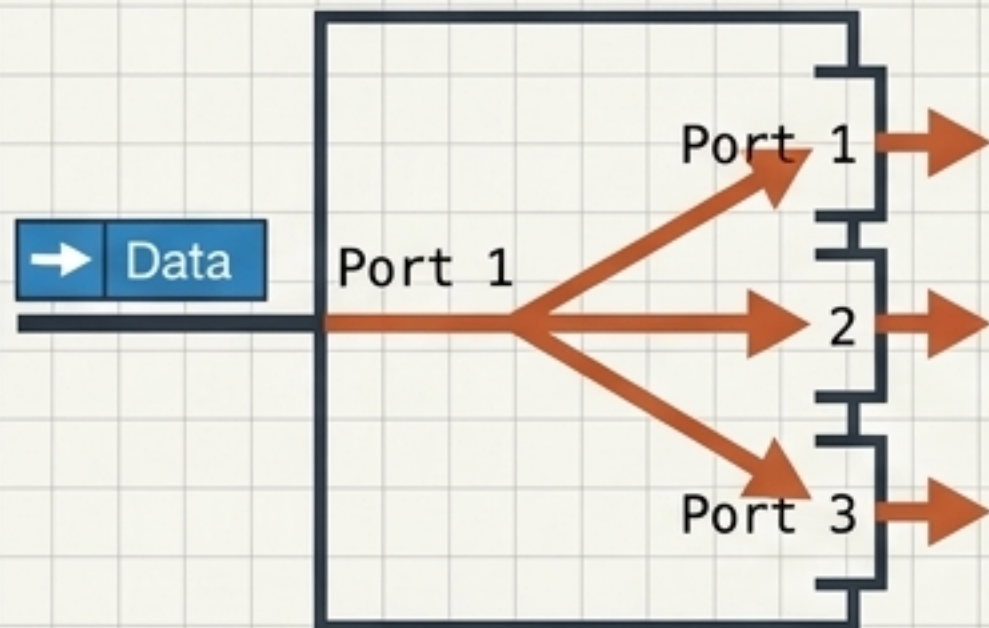
DIAGNOSTICS
Layer: 2
Decision Info: MAC Addresses
Table Maintained: MAC/CAM Table
Domains: Separate collision domains per port
Forwarding Rule: Match destination MAC to port

Switch Forwarding Logic

Unknown Unicast

Condition Box:

CAM table is empty.

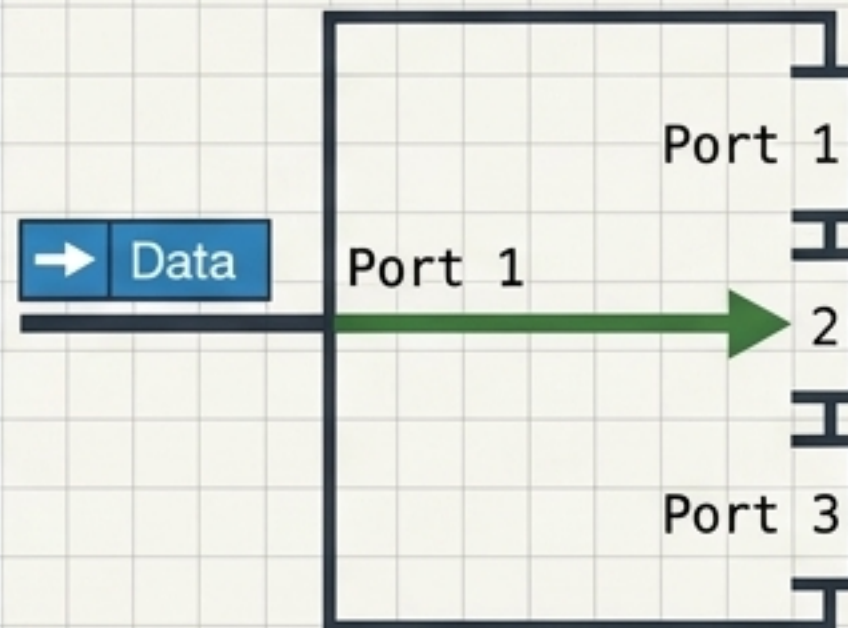


Floods like a hub,
but out of necessity.

Known Unicast

Condition Box:

CAM table has a
matched entry.

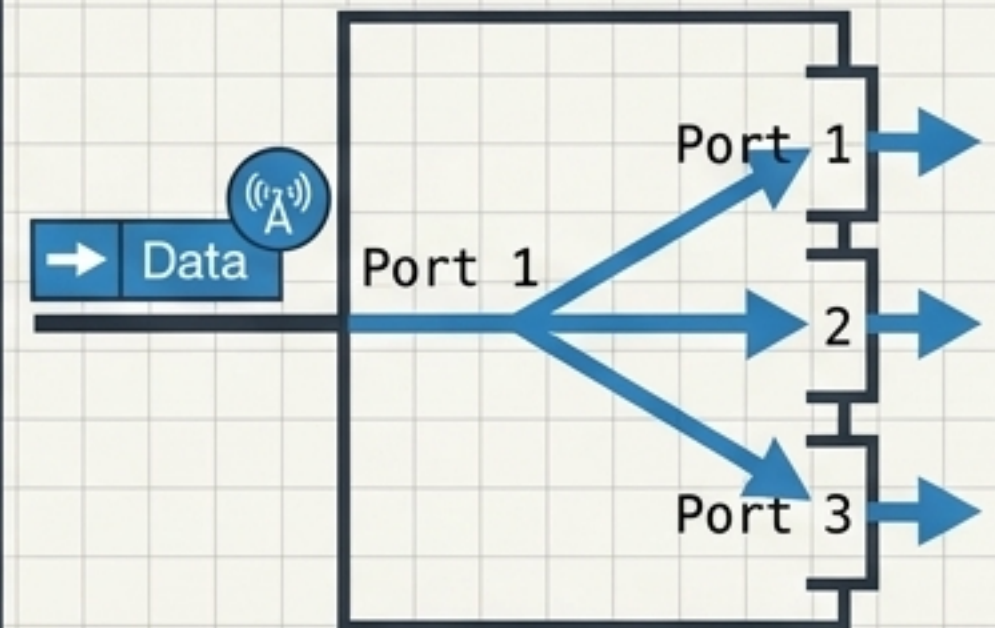


Selective 1-to-1
forwarding.

Subnet Broadcast

Condition Box:

Destination MAC:
FF:FF:FF:FF:FF:FF



Always floods by
design.

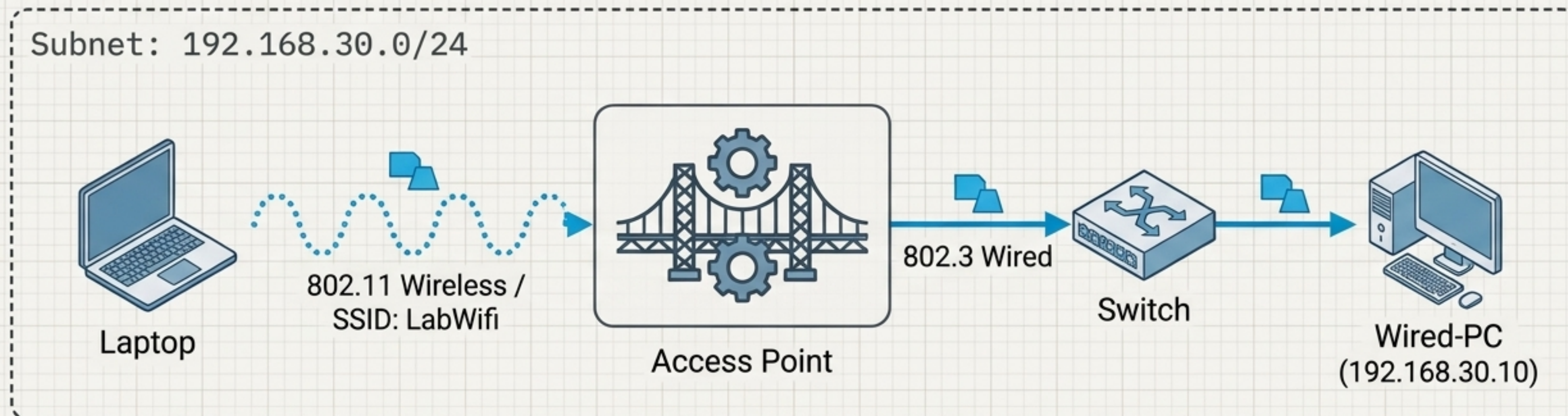
The Layer 1 vs. Layer 2 Divide

Behavior	Hub (Layer 1)	Switch (Layer 2)
First frame to unknown destination	Flood	Flood
Subsequent frames to known destination	Flood	Selective Forward
Broadcast frame	Flood	Flood
Reads destination MAC?	[X] No	[✓] Yes
Learns source MAC?	[X] No	[✓] Yes
Creates separate collision domains?	[X] No	[✓] Yes

This is the true structural upgrade of Layer 2 hardware. Hardware isolation prevents data collisions.

Layer 2: The Access Point (Wireless Bridge)

An Access Point (AP) simply bridges physical mediums. It translates 802.11 wireless frames into 802.3 wired Ethernet frames on the exact same subnet. It performs **zero routing**.

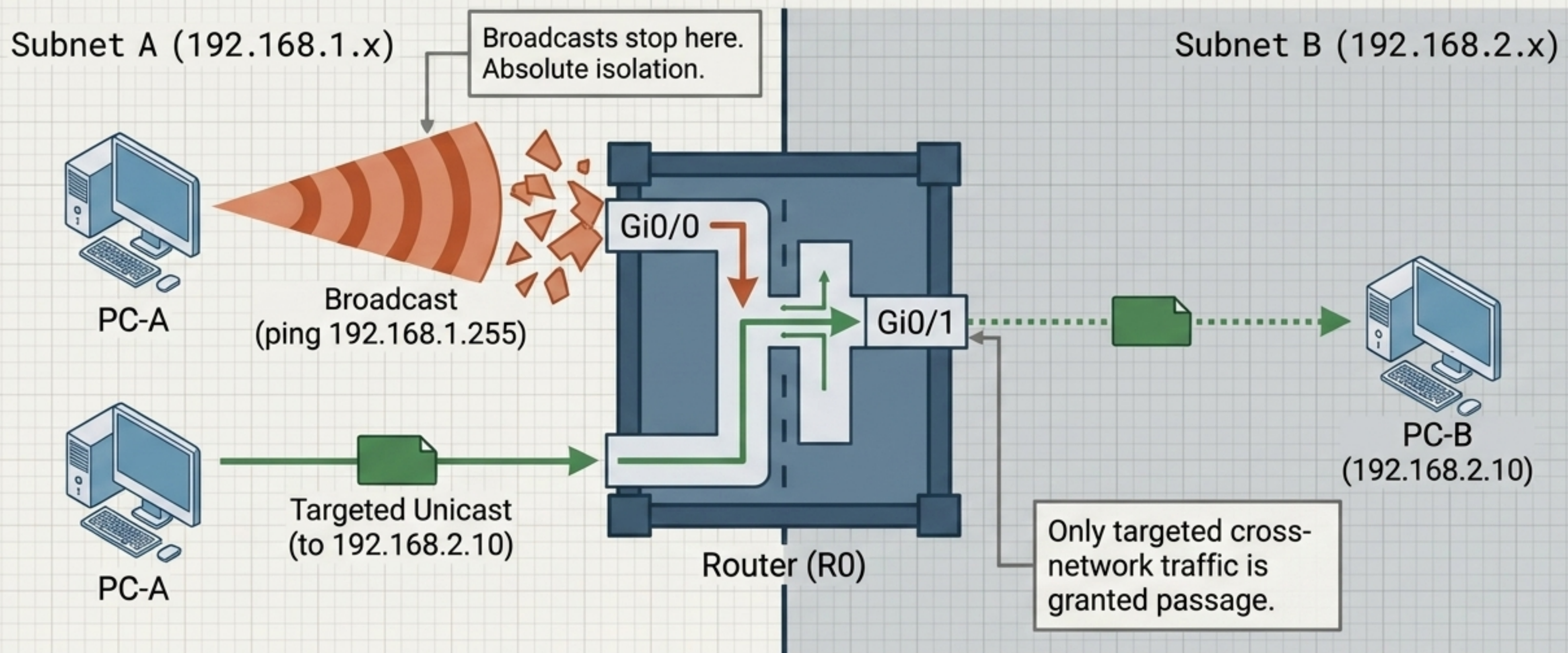


Crucial Limitation:

A cross-subnet ping (e.g., to 192.168.40.10) will fail through an AP alone. A Layer 3 device is required to escape the local network.

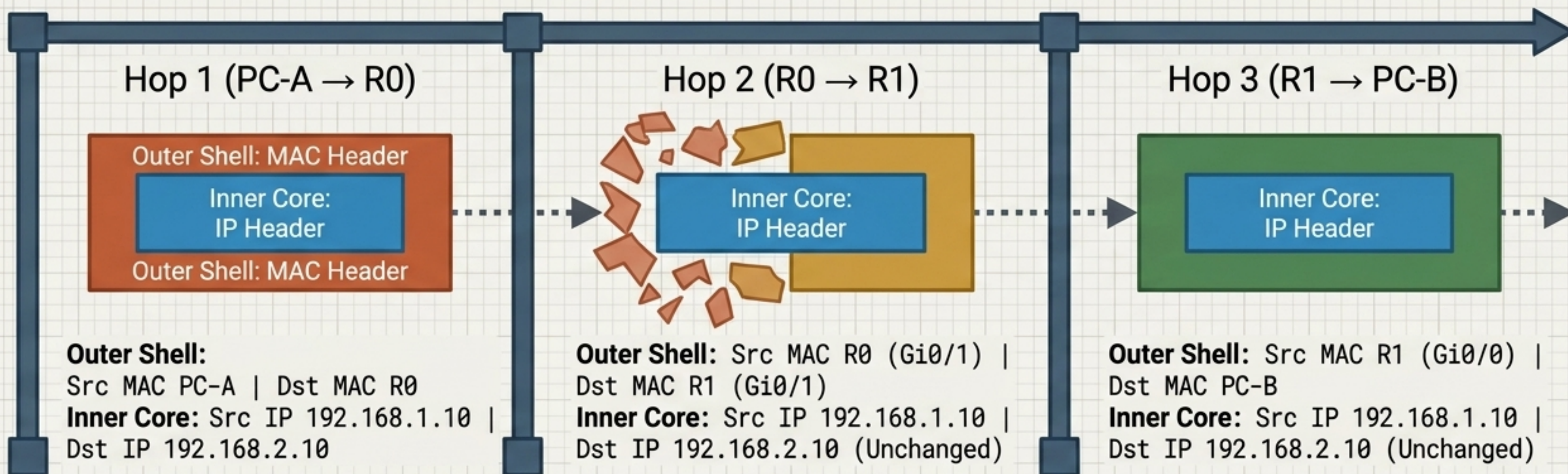
Layer 3: The Router (Logical Mapping)

Traffic within the same subnet flows freely. But different-subnet traffic requires a Router. Routers navigate logical IP maps and act as absolute physical walls against local noise.



The Hop-by-Hop Frame Rebuild

IP is the end-to-end communication. Ethernet is just the local point-to-point delivery vehicle. The outer frame is destroyed and rebuilt at every single router.



Rule: Hosts send traffic to their Gateway's MAC address, **NEVER** the remote destination's MAC.

Routing Logic & The Default Route

```
R0# show ip route
```

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

```
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
```

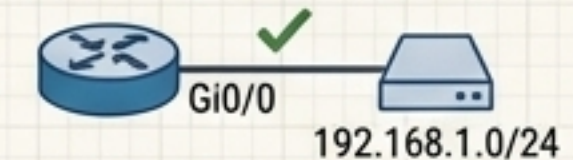
```
C 10.0.0.0/30 is directly connected, GigabitEthernet0/1
```

```
S 192.168.2.0/24 [1/0] via 10.0.0.2
```

```
S* 0.0.0.0/0 [1/0] via 10.0.0.2
```

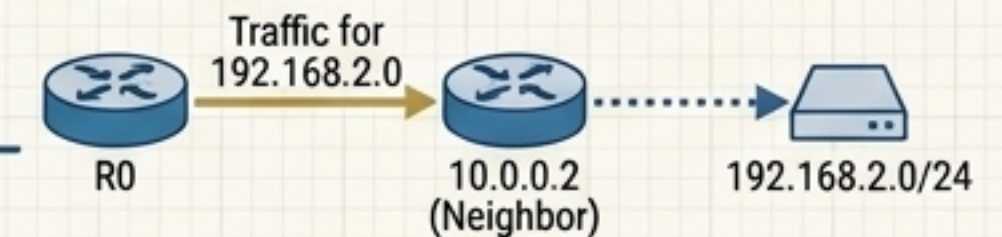
Connected (C)

The router physically touches this network via an active interface. No manual configuration needed.



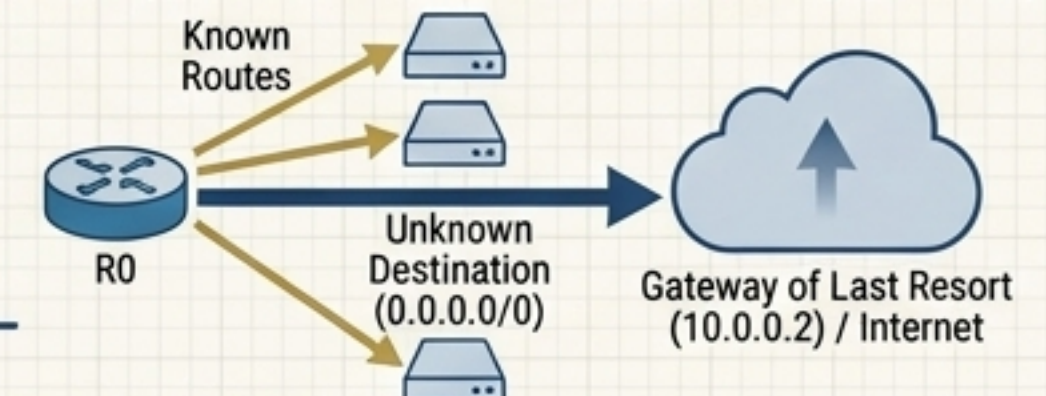
Static (S)

The router was explicitly told by an administrator: "To reach 192.168.2.0, send the traffic out to neighbor 10.0.0.2".



Default Route (S*)

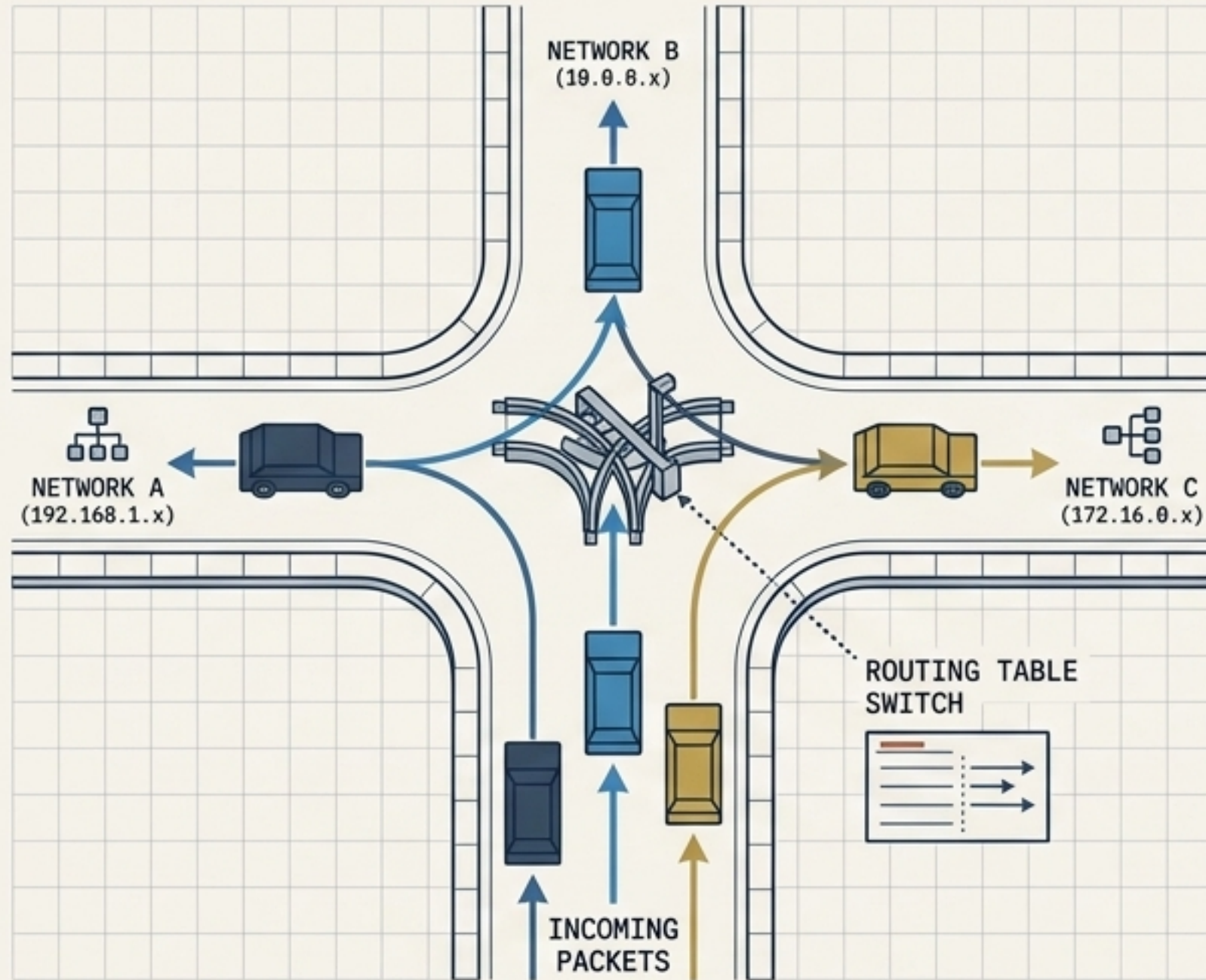
The Gateway of Last Resort. If the destination does not exist anywhere else on the map, blindly send it here. (This is how home routers forward internet traffic).



The Firewall (Security as a Separate Function)

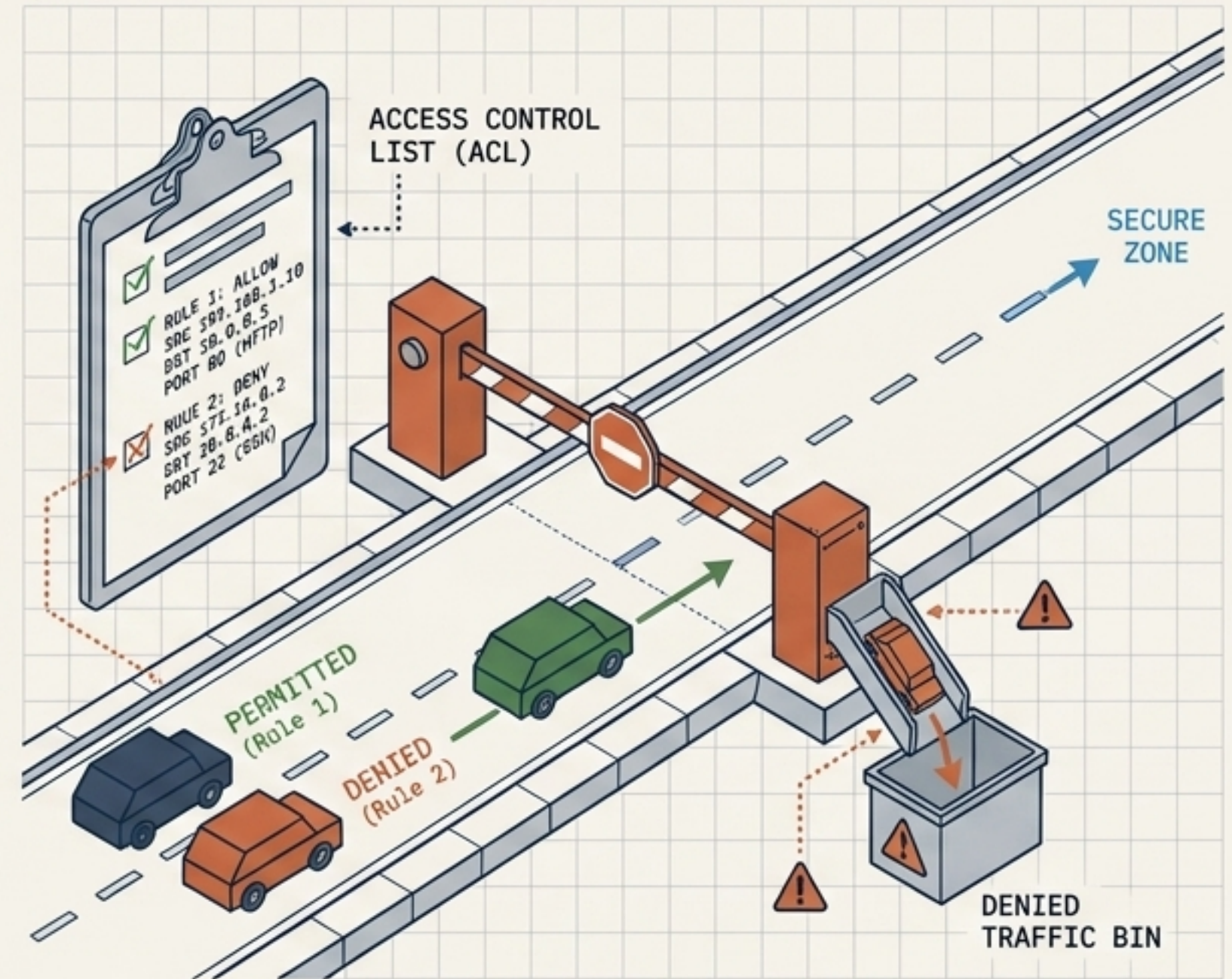
A router asks: Where should this packet go? A firewall asks: Should this traffic be allowed?

The Router: Direction



Navigates paths based on destination mapping.

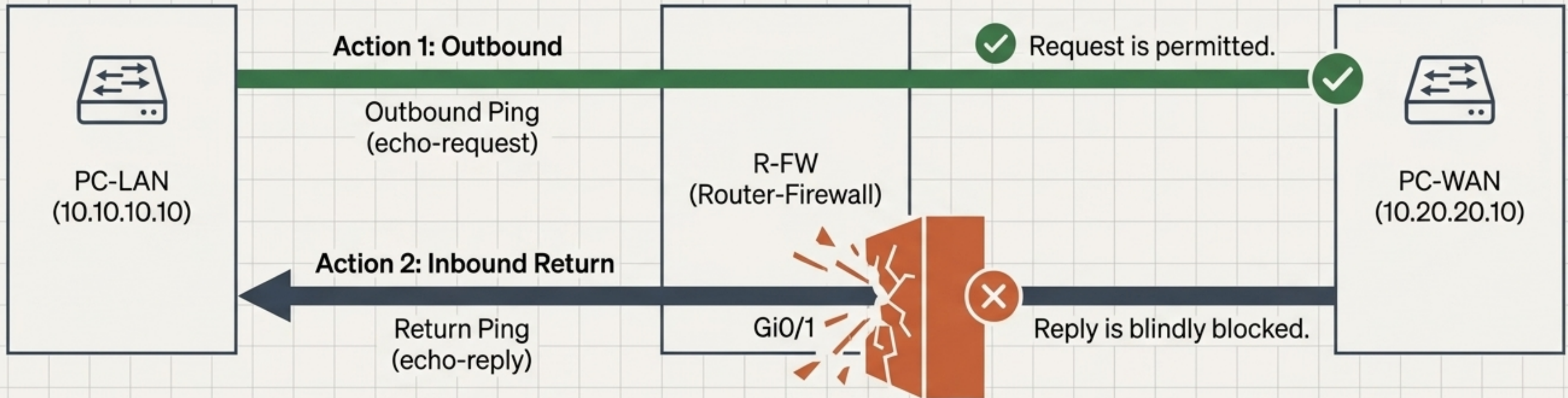
The Firewall: Permission



Inspects and filters traffic based on strict policy rules.

The Trap of Stateless ACLs

```
access-list 101 deny icmp 10.20.20.0 0.0.0.255 any
```

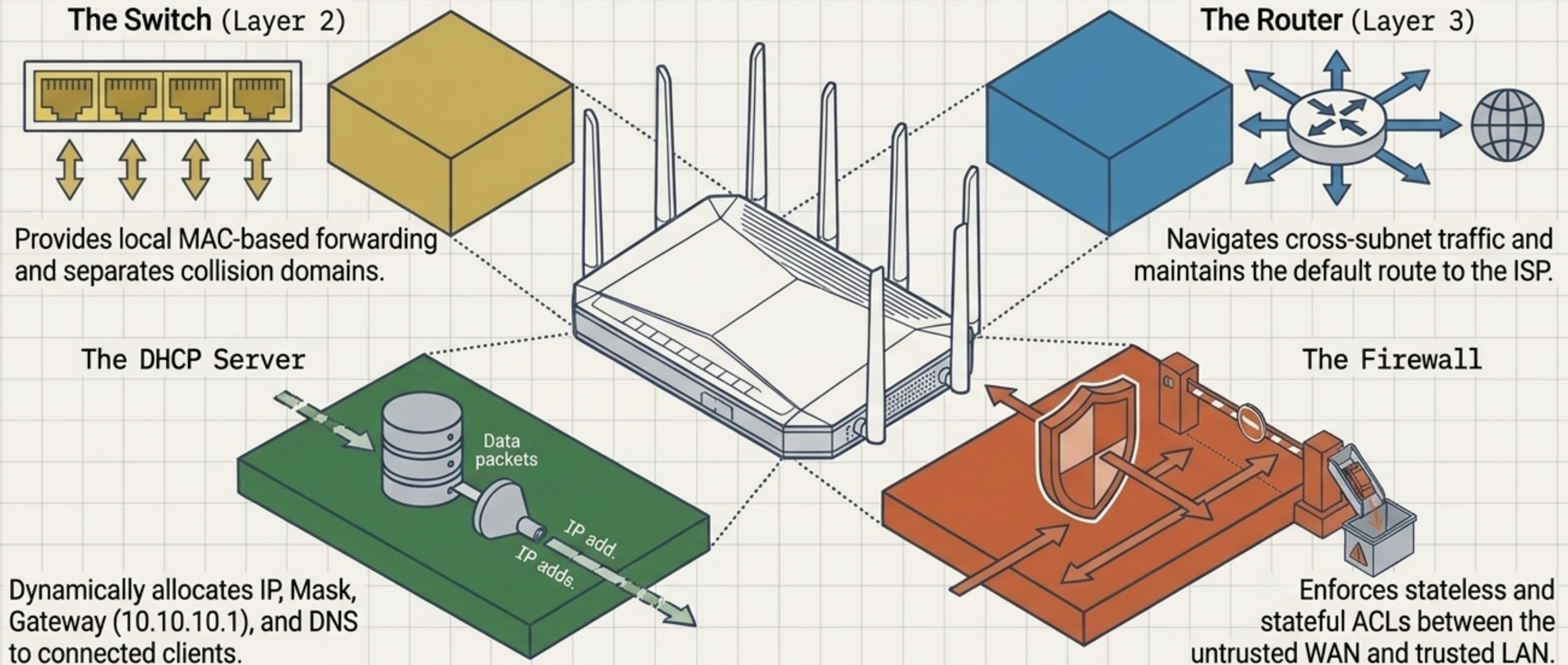


The Stateful Solution:

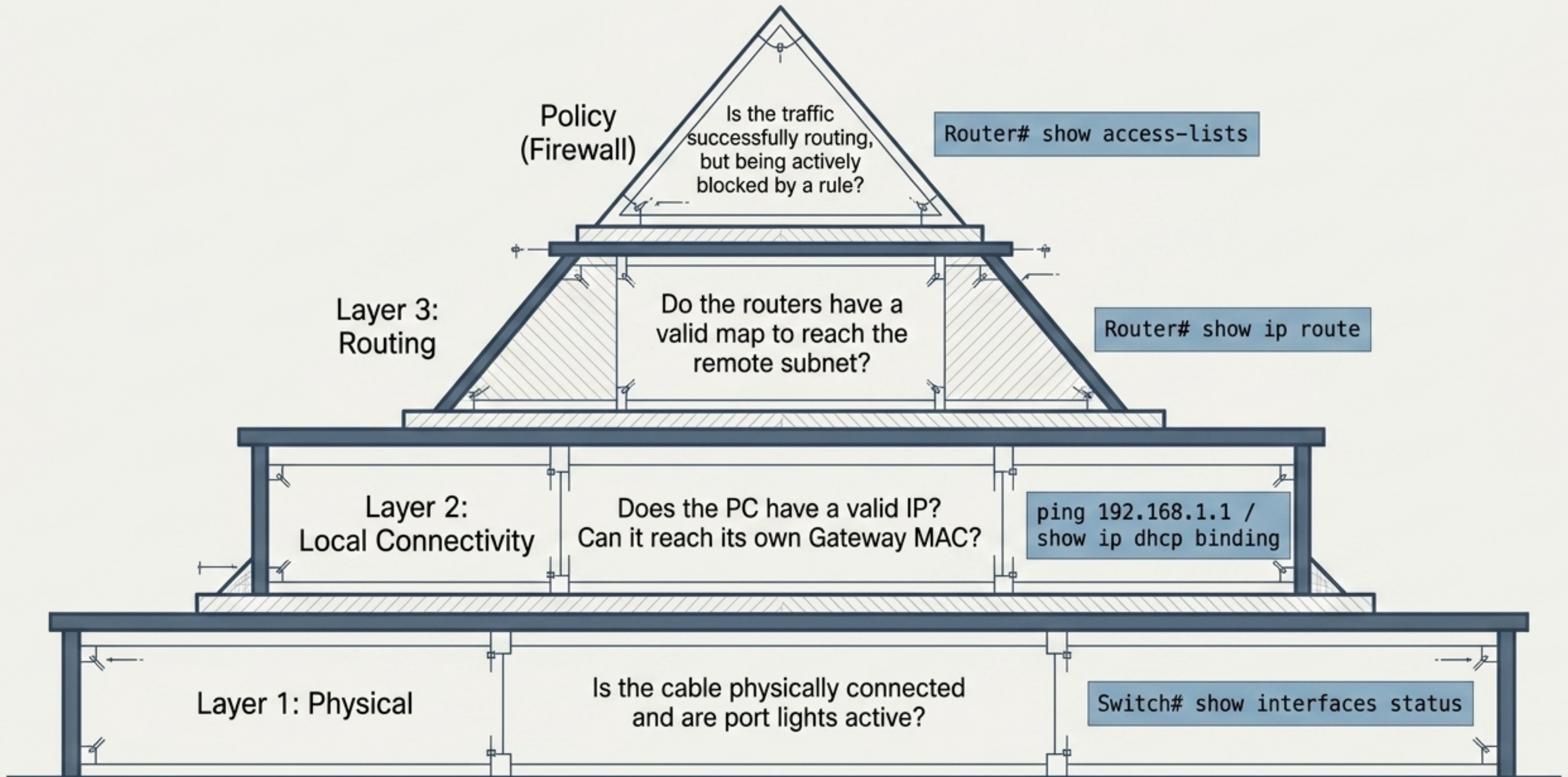
Because this is a stateless ACL, it blindly blocks all ICMP from the WAN—even replies to requests initiated by the trusted LAN. A stateful firewall is required to track active connection states and dynamically allow return traffic.

Synthesis: The Home Gateway

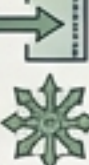
One physical chassis, four distinct logical enterprise functions.
The user sees a magic box; the engineer sees layers of independent intelligence.



The Bottom-Up Troubleshooting Framework



The Master Intelligence Matrix

Device	Layer	Decision Info Used	Table Maintained	Domain Behavior	Forwarding Rule
Hub	1	Physical Signal	None	1 Shared Collision 	Repeat to all 
Switch / AP	2	MAC Address	CAM Table	Separate Collision 	MAC match -> Port (Flood if unknown) 
Router	3	IP Address	Routing Table	Separate Broadcast 	IP match -> Route via map 
Firewall	3+	Policy Rules	ACL / State Table	N/A (Filters)	Match rule -> > Allow/Deny 

Intelligence in networking is simply the progression from physical repetition to logical policy.